

University of Southampton
Southampton, United Kingdom

June 17, 2026

Dear Admission Committee,

As a highly motivated and ambitious AI Data Science Engineer, I am thrilled to apply for the PhD position investigating privacy risks in Large Language Models at the University of Southampton. This project resonates deeply with my research interests and passions, as I aim to leverage my skills in Deep Learning and PyTorch to contribute meaningfully to this interdisciplinary team.

My background in developing innovative AI solutions has equipped me with a strong foundation in mathematical modeling. For instance, during my previous project on Cardiovascular Medical Digital Twin, I designed *MedicalMambaTS—a linear—time statespace model capable of providing sub—millisecond bedside inference for cardiovascular di-*

Furthermore, I have experience working with PyTorch Autograd, which allows me to fine-tune models using gradient-based optimization techniques. This expertise will be valuable in developing novel data mitigation techniques for Large Language Models. My familiarity with PyTorch Geometric has also given me a strong grasp of geometric deep learning concepts, which can be applied to this project's focus on auditing and mitigating privacy risks.

I am particularly drawn to this PhD opportunity because it addresses the pressing issue of privacy and security risks in large language models. As someone who is passionate about developing AI solutions that benefit society, I believe it is essential to explore these risks and develop effective methods for mitigating them. I am excited about the prospect of contributing to a team dedicated to making LLMs-based systems safer, more transparent, and accountable.

Sincerely,

Ismail Aissaoui